

Learning Sub-Grid Physics of Geophysical Turbulence with Fourier Neural Operators: A Case Study in Quasi- Biennial Oscillation Modelling

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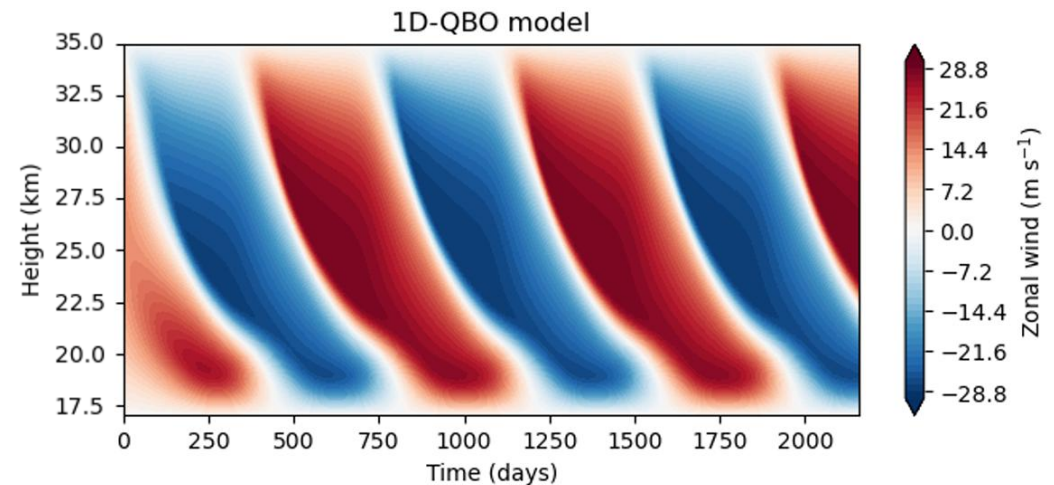
Sub-Grid Parameterization

- Earth system modelling depends on resolution, including ocean and atmosphere;
- Representing sub-grid processes need **parameterization**;
- In climate models, Gravity Wave Parameterization (GWP) is an important component;
- **Quasi-Biennial Oscillation (QBO)** is a phenomenon in stratosphere mainly forced by Gravity Waves.

- QBO can be modelled by (1D-QBO, Plumb [1977]),

$$\frac{\partial \bar{u}}{\partial t} + w \frac{\partial \bar{u}}{\partial z} - \kappa \frac{\partial^2 \bar{u}}{\partial z^2} = \text{GWF}(\bar{u}, z)$$

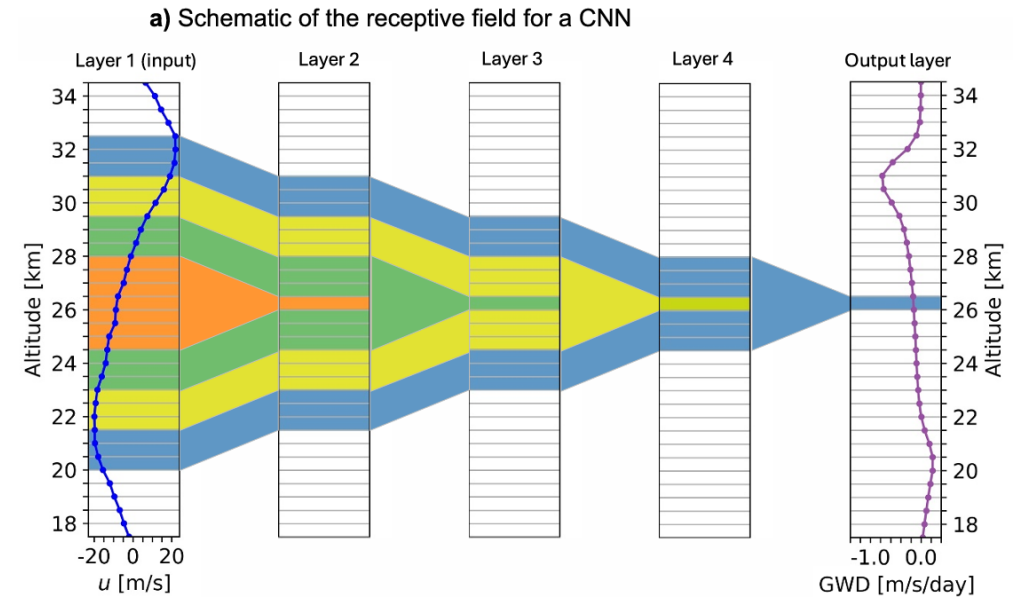
$$\text{GWF}(\bar{u}, z) = -\frac{\rho_L}{\rho_0} \sum_n \frac{\partial}{\partial z} \left[F_L \exp \left\{ - \int_{z_L}^z \frac{N\mu}{k_n [\bar{u}(z) - c_n]^2} dz' \right\} \right]$$



The numerical solution of the 1D-QBO model
(setting: $n = 2, c_1 = -c_2$).

Machine Learning in Sub-Grid Parameterization

- Machine Learning methods have been widely used in Sub-Grid Parameterization:
 - Cloud processes in the climate model (Rasp et al. [2018]);
 - 2D turbulence in the ocean model (Yan et al. [2025]);
 - ...
- Problems:
 - generalization ability (O’Gorman et al. [2018]);
 - physical inconsistency (Beucler et al. [2021]).
- Machine Learning strategies for GWP focus on **CNNs**: sensitive to the receptive fields (Pahlavan et al. [2025]).



Pahlavan et al. [2025], Figure 2a.



Fourier Neural
Operators (**FNOs**)

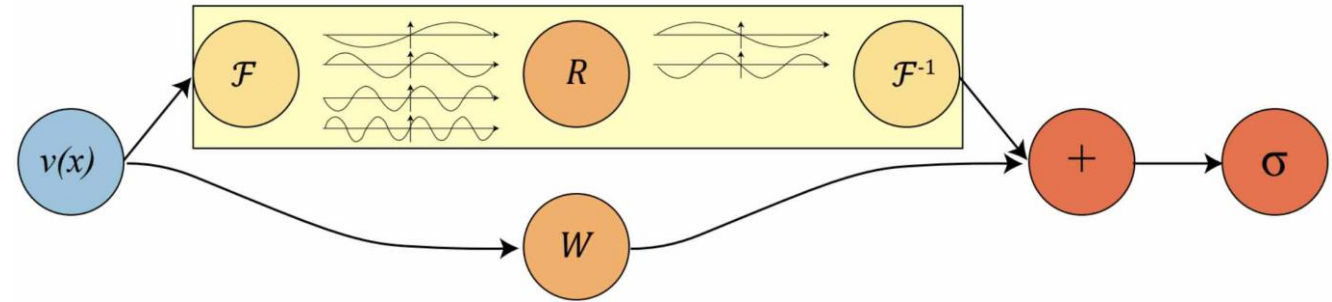
Fourier Neural Operators (FNOs)

- Operator learning:

- Learning map: $\mathcal{G}: v(x) \rightarrow u(x)$
- Resolution-invariance;

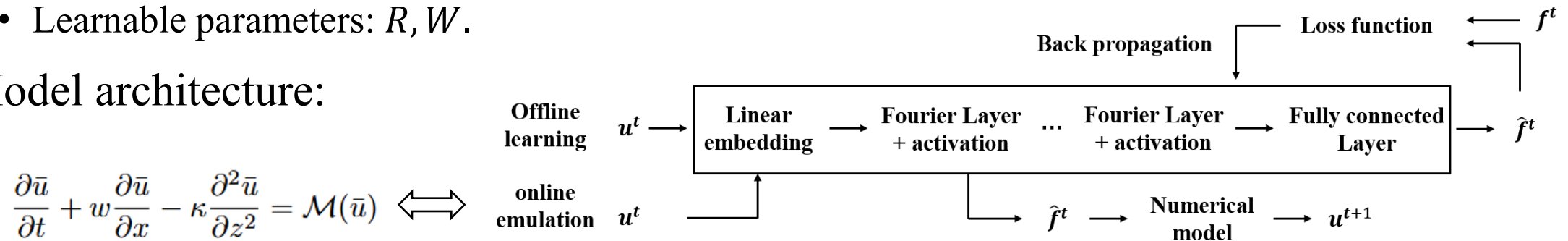
- Fourier layer:

- The **linear transforms** in Fourier space are **convolutions** in real space;
- **mode** cutoff.
- Full domain extent/receptive field;
- Learnable parameters: R, W .



Li et al. [2020], Figure 2b.

- Model architecture:



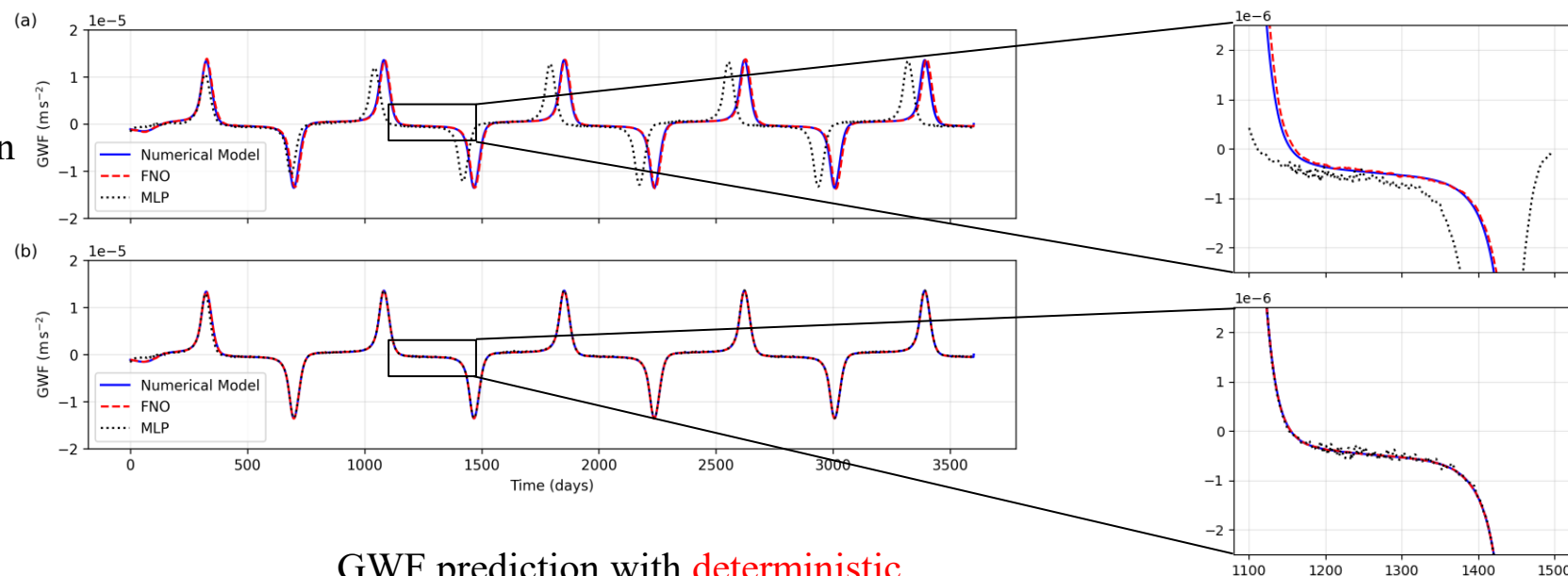
$$\frac{\partial \bar{u}}{\partial t} + w \frac{\partial \bar{u}}{\partial x} - \kappa \frac{\partial^2 \bar{u}}{\partial z^2} = \mathcal{M}(\bar{u}) \iff$$

- Comparison: MLP with a very large parameter size (> 2 million).

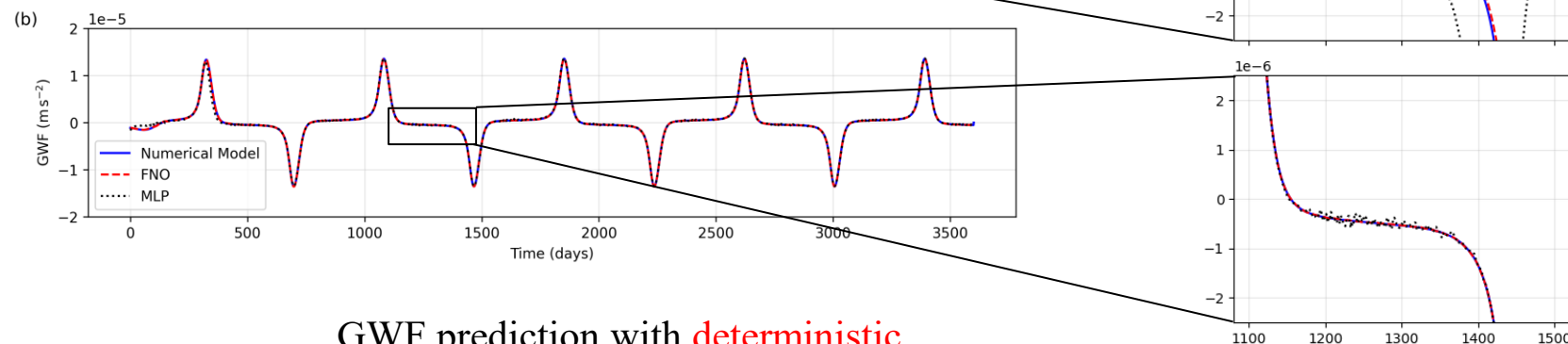
Results

Reproduced from Pahlavan et al., *On the Importance of Learning Non-Local Dynamics for Stable Data-Driven Climate Modeling: A 1D Gravity Wave QBO Testbed*

Online emulation

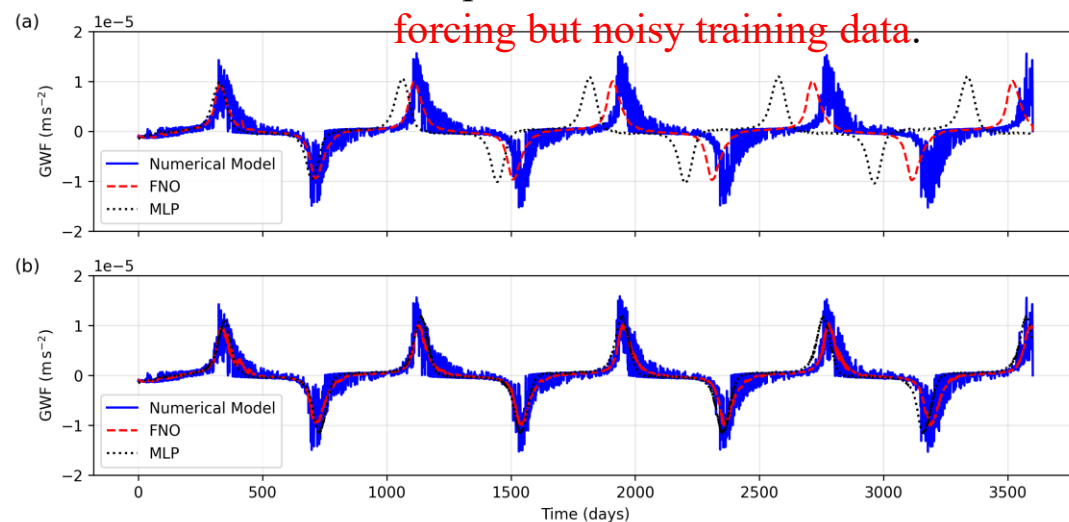


Offline testing

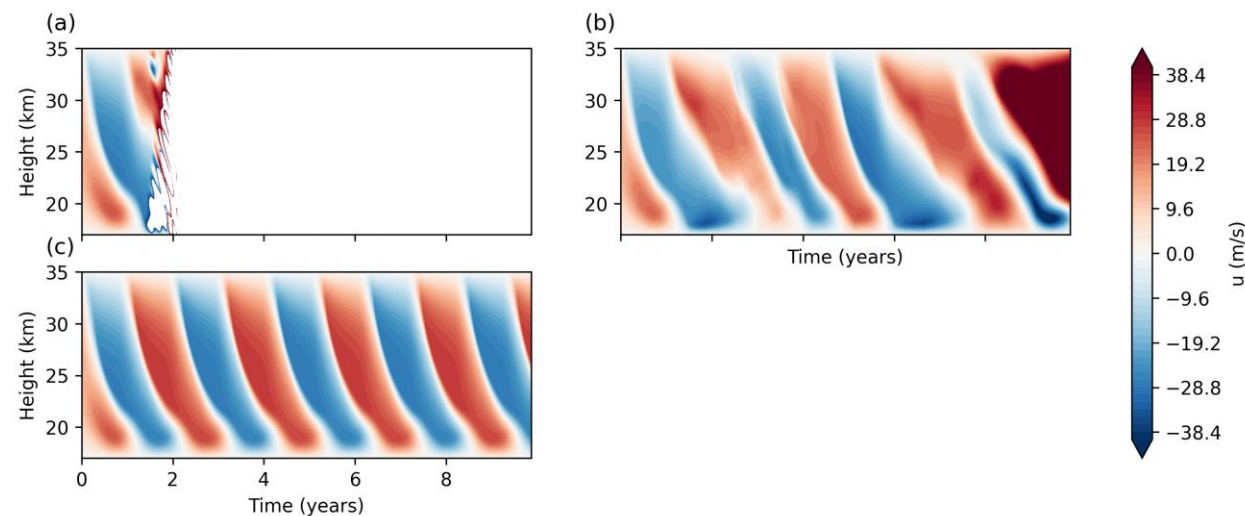


There are some small fluctuations in the prediction of MLP.

GWF prediction with **deterministic forcing but noisy training data.**



GWF prediction with **stochastic forcing.**



Online performances of the models with different complexity.

Conclusions and Discussions

- An FNO with a very small parameter size can simulate the QBO period more accurately than a deep MLP in this task;
- FNOs still have difficulty learning the wave forcing with a wide spectrum.
- Technical improvement: hyperparameter tuning...
- Physical consistency:
 - Probability density functions (Pahlavan et al. [2024]);
 - Energy conservation;
 - PINOs (Li et al. [2024]).
- FNOs' structure:
 - Act as a lowpass filter;
 - Resolution-invariance.
- Implemented into GCMs.